

optics in software. Such a capability allows use of simpler optics and enables new optical architectures. The optical correction capability of holograms allows arbitrary optical corrections on a per-pixel basis.

**Form factor.** One of the most important considerations for a near-eye display is form factor, especially for see-through, mixed-reality devices. Lightweight, eyeglasses-like displays are needed to facilitate viewer comfort and all-day use.

### Technological developments

In the past, researchers interested in holographic display systems proposed or focused on methods for overcoming limitations in the combined spatial resolution and speed of commercially available, spatial light modulators. Representative techniques included space-division multiplexing, time-division multiplexing and combination of these two techniques.

Implementing a viewing window design demands close attention to the optical image system. Holography is being advanced through development of structure materials at the nanoscale. These allow devices to achieve new optical properties that go beyond the properties of natural materials.

With a tabletop display, a viewing window can be created by using a magnified virtual hologram, but the plane of the image is tilted with respect to the rotational axis and projected using two parabolic mirrors. As parabolic mirrors do not have an optically-flat surface, visual distortion can result.

This problem can be solved by designing an aspheric lens, which allows multiple viewers to observe 8cm (3.2-inch) holograms from any position around the table without visual distortion. Building on these advances, researchers hope to implement a key design feature of strategically sizing the viewing window, so that it is closely related to the effective pixel size of the rotating image of the virtual hologram. Watching through this window, observers' eyes are positioned to accept the holographic image light field because

the system tilts the virtual hologram plane relative to the rotational axis.

To enhance the viewing experience, the team hopes to design a system in which observers can see 8cm holographic 3D images floating on the surface of the parabolic mirror system at a rate of 20 frames per second. Test results of the system using a 3D model and computer-generated holograms were promising—though right now still in a monochrome green colour. Next, the team wants to produce a full-colour experience and resolve issues like undesirable aberration and brightness mismatch among the four digital micromirror devices used in the display.

In an attempt to assimilate holographic technology with modern electrical devices, a team of researchers has created the world's thinnest hologram. The team claims the thickness of holograms has typically been limited to the optical wavelength scale, something which seems to have reduced integration with thin electronic devices. However, researchers were able to surpass this limit and achieve holograms 60nm in size using a topological insulator material—a material which seems to behave as an insulator while also having conducting states upon its surface.

An advantage of this new nano-hologram is the way users may observe holographic projections outside of 3D glasses requirements. This means consumers may potentially witness nano-holograms built into electronic devices such as smartphones, computers or TVs in the future as they require additional peripherals necessary for operation. With this breakthrough in technology, the research team claims the next step for the technology is the development of a new method to adapt holography for LCD screens.

A graduate student at MIT (USA) Media Lab has developed an inexpensive optical chip that, when combined with a prototype display, "renders colour holographic video at resolutions equal to that of a standard-definition TV. Besides the low cost of the chip, the true advance

is the way the colours are handled. In the past, red, green and blue light had to be handled separately, resulting in lower image resolution. This chip handles these lights simultaneously, allowing for both higher resolutions and more efficient processing."

The new approach to generating holograms could lead to colour holographic-video displays that are much cheaper to manufacture than today's experimental (2D), monochromatic displays. The same technique could also increase the resolution of conventional 2D displays. Thus, in the original *Star Wars*' movie, the inviting but grainy special effects hologram might soon be a true full-colour, full-size holographic image due to advances by a research team refining 3D holographic displays. The team described a novel tabletop display system that allows multiple viewers to simultaneously view a hologram showing a full 3D image as they walk around the tabletop, giving complete 360-degree access.

A South Korean research team has developed a 3D holographic display that performs more than 2600 times better than existing 3D holographic displays. It is expected to improve the limited size and viewing angle of 3D images, which were a major problem of the current holographic displays.

To create a 3D hologram that can be viewed without special equipment such as 3D glasses, the wavefront of light must be controlled using wavefront modulators such as spatial light modulators and deformable mirrors. A wavefront modulator is an optical manipulation device that can control the direction of light propagation. However, the biggest limitation to using these modulators as 3D displays is the number of pixels.

The large numbers of pixels that are packed into high-resolution displays developed in recent years are suitable for a 2D image, and the amount of information contained in those pixels cannot produce a 3D image. For this reason, a 3D image that can be made with existing wavefront modulator technology is 1cm in size with a narrow viewing angle of 3 degrees, which is far from practicable. **EFY**